

Attention-based Pairwise Semantic Fusion for Youtube Clickbait detection

ljfiS

Abstract

As the popularity of YouTube continues to grow, competition for attracting views has intensified, leading to a rapid increase in clickbait content that induces user clicks through misleading titles and thumbnails. As a result, users frequently encounter videos whose actual content does not align with the expectations formed by the title and thumbnail. To address this issue, this study proposes a multimodal clickbait detection framework that integrates linguistic information (video titles and automatic speech recognition (ASR) transcripts) and visual information (thumbnails and keyframes). Visual information is first converted into textual representations through image captioning, and all textual data are embedded using Sentence-BERT (SBERT). Subsequently, modality-specific pairs are constructed, and attention pooling is applied to each pair to capture the semantic relevance between expected information and actual content. Experimental results demonstrate that the proposed approach improves clickbait detection by effectively capturing semantic relationships between titles, thumbnails, and actual video content.

Keywords: Clickbait Detection, Semantic Inconsistency, Multimodal, Attention pooling

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*Correspondence to:

()

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1. Introduction

Recently, online video platforms such as YouTube have become one of the most influential media for information consumption and communication worldwide. As these platforms have rapidly grown, competition for views and advertising revenue has intensified, leading to the widespread adoption of various forms of clickbait. Clickbait refers to strategies that induce clicks through sensational or exaggerated title and thumbnail. While such practices may temporarily increase view counts, they ultimately undermine content credibility, induce user fatigue, and may result in user disengagement from the platform. Against this backdrop, understanding and detecting the semantic inconsistency between titles or thumbnails and the actual video content has emerged as an important research problem. The negative perception users experience when encountering clickbait does not primarily stem from exaggeration or emotional expression itself. Instead, the core issue lies in the semantic mismatch between the expectations formed by the title or thumbnail and the content delivered by the video.

However, existing studies have largely defined clickbait in terms of stylistic linguistic features such as exaggerated expressions, emotional language, sensational wording, or references to celebrities, and have focused on detecting these characteristics [1,2]. Since such stylistic elements are also commonly used in general content to attract viewer interest, relying solely on the presence of specific expressions is insufficient for reliably determining content distortion. This expression-centered perspective has introduced methodological bias in clickbait detection. Prior research has tended to rely excessively on linguistic signals such as titles or

ASR transcripts [3], while relatively neglecting visual information such as thumbnails or video frames. In addition, several studies have attempted to utilize statistical metadata such as view counts or like-dislike ratios; [4,5] however, such indicators are only available after content consumption, which makes them unsuitable for preventive detection at the time of upload. Although recent studies have explored multimodal approaches that jointly consider text and images, they still exhibit limitations in capturing meaningful semantic relationships across modalities. Existing methods typically encode text and images independently and then fuse them using element-wise averaging [5]. This strategy fails to precisely model the subtle semantic conflicts between the “expected information” conveyed by titles or thumbnails and the “actual information” presented in the video content. Consequently, the semantic gap between expectations and reality is not effectively captured.

To address these limitations, we propose a multimodal framework that leverages four modalities: title, thumbnail, transcript, and keyframe. By combining these modalities, we construct four semantic comparison pairs: title-transcript, title-keyframe, thumbnail-transcript, and thumbnail-keyframe. Each pair is designed to directly extract semantic relationships between modalities. The resulting representations are integrated through a branch-level attention mechanism, enabling the model to learn the relative importance of each modality pair. This process produces a unified representation that reflects which modality interactions contribute most to clickbait detection, and this representation is used for final prediction.

Experiment results show that our framework effectively captures semantic relationships between titles, thumbnails, and actual video content.

The main contributions of this work are summarized as follows:

- We propose addressing **clickbait detection** from the perspective of semantic incongruity, emphasizing the relationship between expectations formed by titles and thumbnails and the actual video content.
- Second, we present a multimodal architecture that systematically constructs **four semantic comparison pairs** using title, thumbnail, transcript, and keyframe modalities.
- Finally, we introduce a **branch-level attention-based integration mechanism** that captures fine-grained semantic interactions across modality pairs and effectively aggregates critical information for clickbait prediction.

2. Data Construction

This study constructed a multimodal dataset based on news videos to quantitatively analyze the semantic relationships among titles, thumbnails, transcripts, and keyframes.

2.1 Data Collection and Artificial Clickbait Generation

We collected short news videos of approximately three minutes in length from the official YouTube channels of major Korean broadcasters, including SBS¹, KBS², MBC³, and YTN⁴. The collected videos were limited to content such as celebrity news, social issues, and interviews, which contain rich visual and linguistic cues suitable for evaluating semantic inconsistency-based clickbait detection. Samples with poor audio quality, low resolution, or duplicated titles or thumbnails were filtered out in advance. Collecting enough real-world clickbait videos exhibiting semantic inconsistency is challenging because such videos are relatively rare, may be quickly removed due to platform policies, and require human reviewers to inspect the full video content. To overcome these limitations, this study adopts a strategy of artificially generating clickbait samples. Specifically, non-clickbait videos with similar topics are paired, and intentional semantic inconsistencies are introduced by swapping titles and thumbnails within each pair. The pairing was manually performed based on human judgment of topical similarity; for example, videos (0, 1), (2, 3), and (4, 5) were grouped as topically similar pairs. For each video v_i , the title, thumbnail, transcript, and keyframes were extracted.

The transcript generated via automatic speech recognition is hereafter referred to as STT (Speech-to-Text). Prior to embedding, the original non-clickbait videos constituting each pair (v_{2i}, v_{2i+1}) are defined as follows: $v_{2i} = (\text{title}_{2i}, \text{thumbnail}_{2i}, \text{STT}_{2i}, \text{keyframe}_{2i})$ and $v_{2i+1} = (\text{title}_{2i+1}, \text{thumbnail}_{2i+1}, \text{STT}_{2i+1}, \text{keyframe}_{2i+1})$. Artificial clickbait samples were generated as $v_{2i} = (\text{title}_{2i+1}, \text{thumbnail}_{2i+1}, \text{STT}_{2i}, \text{keyframe}_{2i})$ or $v_{2i+1} = (\text{title}_{2i}, \text{thumbnail}_{2i}, \text{STT}_{2i+1}, \text{keyframe}_{2i+1})$. In this process, the actual video content namely the STT and keyframes remains unchanged, while only the expectation-setting elements (titles and thumbnails) are replaced. This procedure introduces intentional semantic inconsistency between the expected information and the actual video content without modifying the underlying video itself. In addition to the

¹ <https://www.youtube.com/@sbsnews8>

² <https://www.youtube.com/@newskbs>

³ <https://www.youtube.com/@MBCNEWS11>

⁴ <https://www.youtube.com/@ytinstar>

Dataset Construction Pipeline and Example Samples

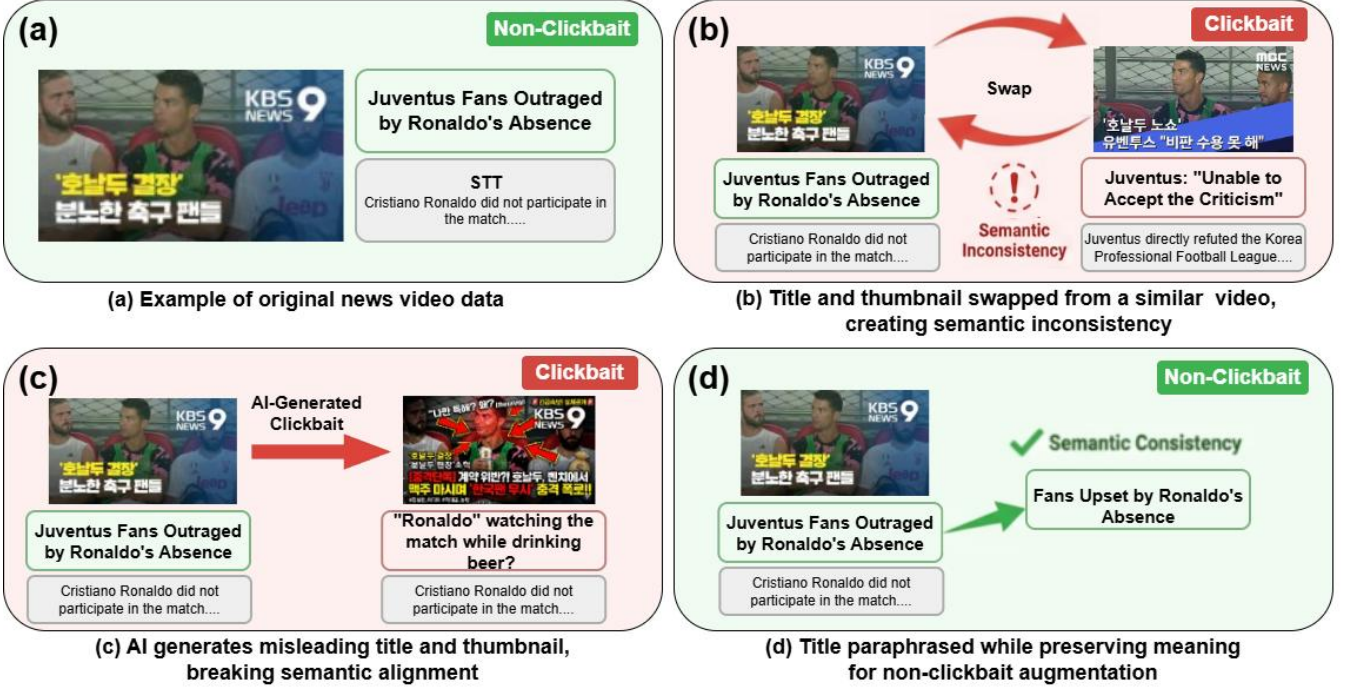


Figure 1. Example of Dataset Construction Methods

swap-based construction, we employed an AI-based generation strategy to introduce more diverse forms of semantic inconsistency. GPT-5 was provided with the original video title and STT and prompted to generate a clickbait-style title that was semantically inconsistent with the actual video content. Based on the generated title, Gemini Nano Banana 2 was then used to generate a corresponding clickbait thumbnail aligned with the misleading title. This process complements simple modality swapping by introducing semantic inconsistencies generated through generative models.

For non-clickbait samples, we aimed to prevent the model from relying on superficial correlations between specific lexical patterns and clickbait labels. To this end, the original titles were paraphrased while preserving their semantic meaning. This approach maintains semantic consistency while increasing linguistic diversity, thereby encouraging the model to focus on semantic alignment between the expectation-setting information and the actual video content rather than on surface-level lexical patterns. Figure 1 illustrates the overall dataset construction process, including both swap-based augmentation and AI-based clickbait generation. The detailed dataset composition and experimental splits are provided in Section 4.1.

shared semantic space, we unified all modality representations into a text-based format.

a) Thumbnail captioning

For each video, the thumbnail image was captioned using the Qwen2-VL-7B-Instruct model [6] together with the corresponding title, converting visual information into a textual representation.

b) Title preprocessing

Video titles were refined by removing channel-specific tokens (e.g., broadcaster names) and unnecessary special characters to normalize the textual representation.

c) STT generation

The audio signal of each video was converted into text using the Whisper ASR model [7]. Through these preprocessing steps, thumbnails, titles, and transcripts were unified as text-based inputs, allowing them to be compared and integrated within a shared semantic space during the embedding stage.

d) Keyframe extraction and scene captioning

To structurally represent video content, a four-stage procedure was applied to extract keyframes at the scene level and convert t

2.2 Modality-specific Feature Extraction and Textual Unification

To enable comparison and learning across all modalities within a

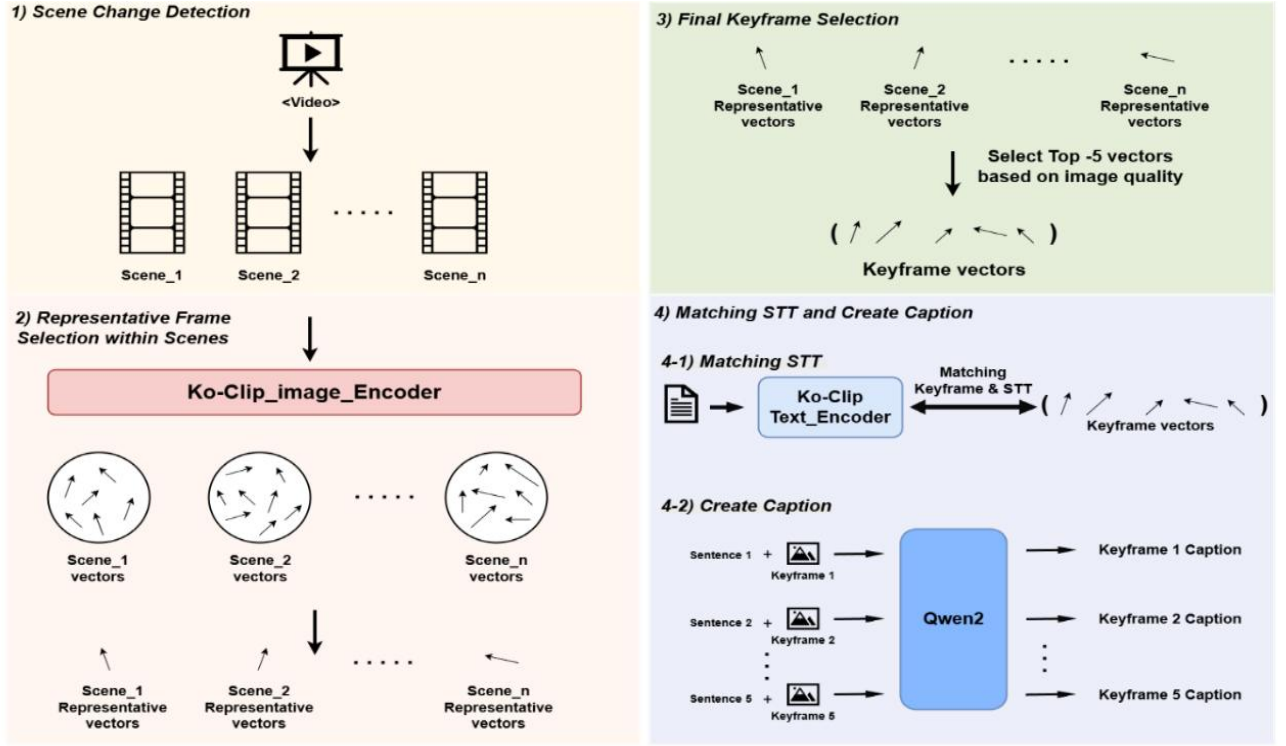


Figure 2. Overall pipeline for generating scene-level captions from video

hem into textual descriptions.

Scene change detection: Scene boundaries were defined based on abrupt differences in color histograms and pixel variations between consecutive frames. Since scene boundaries correspond to points of visual change in the video, selecting keyframes based on these boundaries reduces redundancy while preserving representative visual information, making this approach well suited for semantic inconsistency analysis.

Representative frame selection within scenes: Within each scene segment, frames were embedded using the KoCLIP image encoder [8]. The frame whose embedding was closest to the mean embedding of the segment was selected as the representative frame for that scene.

Final keyframe selection: Among the representative frame candidates, the top five frames with the highest image quality scores were selected as the final keyframes.

Matching STT and caption generation: Cosine similarity was computed between each extracted keyframe image and the STT sentences, and the most semantically relevant STT sentence was matched to each keyframe. These matched pairs were then provided as input to the Qwen2-VL-7B-Instruct model to generate visual cues with linguistic context.

Figure 2 illustrates the overall keyframe captioning pipeline, including scene change detection, keyframe selection, STT

matching, and caption generation. Through this process, titles, audio, thumbnails, and keyframes were all converted into textual representations. Finally, all textual inputs were embedded using Sentence-BERT.

3. Multimodal Framework for Semantic Consistency

3.1 Overview

We propose a novel multimodal model that directly compares the semantic consistency between the expectations presented by video titles and thumbnail caption and the actual video content, represented by STT and keyframe captions. The proposed model embeds four input modalities—title, thumbnail caption, STT and keyframe captions—into a shared semantic space and then employs an attention-based mechanism to extract semantically important segments from the video content (STT and keyframe captions) with respect to the title and thumbnail caption.

Subsequently, relational representations are constructed for each of the four modality pairs, and these signals are integrated through branch-level attention to ultimately determine whether the video exhibits clickbait characteristics. Figure 3 illustrates the overall architecture of the proposed framework.

3.2 Attention Pooling and Pairwise Relation Modeling

The proposed model utilizes four input modalities—title, thumbnail caption, STT and keyframe captions. Each input is

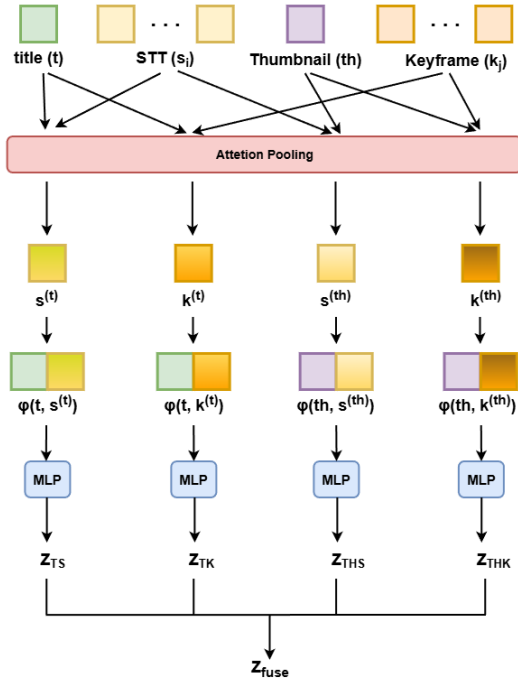


Figure 3. Overview of the proposed Pairwise-Semantic Fusion architecture

transformed into a 768-dimensional embedding vector ($\mathbf{h} \in \mathbb{R}^{768}$) using a Sentence-BERT (SBERT) model, and L2 normalization is applied to ensure stability in cosine similarity computation. The title and thumbnail caption are represented as single vectors ($\mathbf{t}$, $\mathbf{th}$), whereas the STT and keyframe captions are represented as variable-length sequences ($\{\mathbf{s}_i\}$, $\{\mathbf{k}_j\}$).

Attention Pooling

Since the STT and keyframe captions consist of multiple tokens, an attention-pooling mechanism is employed to aggregate the token embeddings into a single vector [9]. In this process, the title ($\mathbf{t}$) or thumbnail ($\mathbf{th}$) embedding is used as the query. Here $\mathbf{x}_i$ denotes a generic token embedding from the input sequence, which corresponds to either an STT token ($\mathbf{s}_i$) or a keyframe caption token ($\mathbf{k}_j$), depending on the modality. For each sequence token $\mathbf{x}_i$ and query $\mathbf{q} \in \{\mathbf{t}, \mathbf{th}\}$, a cosine similarity score u_i is computed, and the corresponding attention weight α_i is obtained by applying SoftMax normalization, as defined in the following equation [9].

$$u_i = \frac{\cos(\mathbf{x}_i, \mathbf{q})}{\tau}, \alpha_i = \frac{\exp(u_i)}{\sum_j \exp(u_j)} \quad (1)$$

where τ is a temperature parameter controlling the sharpness of the distribution. The aggregated vector is then obtained as

$$\bar{\mathbf{x}} = \sum_i \alpha_i \mathbf{x}_i \quad (2)$$

Through this process, the model uses the title or thumbnail caption as a query and captures the semantic relevance to the corresponding STT sentences and keyframe captions sequences of the video content, thereby transforming variable-length sequences into fixed-length semantic vectors.

Pairwise Feature Construction

To explicitly capture relational semantics across modalities, two embeddings $\mathbf{h}_1$ and $\mathbf{h}_2$ are combined to form a pairwise feature vector [10]:

$$\boldsymbol{\varphi}(\mathbf{h}_1, \mathbf{h}_2) = [\mathbf{h}_1; \mathbf{h}_2; |\mathbf{h}_1 - \mathbf{h}_2|; \mathbf{h}_1 \odot \mathbf{h}_2] \in \mathbb{R}^{4d} \quad (3)$$

This formulation encodes both the semantic difference between the two embeddings ($|\mathbf{h}_1 - \mathbf{h}_2|$) and their element-wise interactions ($\mathbf{h}_1 \odot \mathbf{h}_2$). In total, four pairwise branch representations are constructed: $\boldsymbol{\varphi}(\mathbf{t}, \mathbf{s}^{(t)})$ for title-STT, $\boldsymbol{\varphi}(\mathbf{t}, \mathbf{k}^{(t)})$ for title-keyframe captions, $\boldsymbol{\varphi}(\mathbf{th}, \mathbf{s}^{(th)})$ for thumbnail caption-STT, and $\boldsymbol{\varphi}(\mathbf{th}, \mathbf{k}^{(th)})$ for the thumbnail caption-keyframe captions relationships. All branch inputs are normalized using Layer Normalization to ensure numerical stability.

3.3 Branch-Level Attention Fusion

The outputs of the four previously generated branches are combined into a single matrix $\mathbf{Z}$:

$$\mathbf{Z} = [\mathbf{z}_{TS}, \mathbf{z}_{TK}, \mathbf{z}_{THS}, \mathbf{z}_{THK}] \in \mathbb{R}^{4 \times 4d} \quad (4)$$

A learnable attention weight matrix $\mathbf{W}_a \in \mathbb{R}^{4d \times 1}$ is applied to compute the contribution of each branch, and the resulting scores are normalized using a SoftMax function to obtain attention weights α_b .

$$\alpha_b = \frac{\exp(\mathbf{W}_a^T \mathbf{z}_b)}{\sum_{b'} \exp(\mathbf{W}_a^T \mathbf{z}_{b'})} \quad (5)$$

The final fused representation is defined as the weighted sum of the branch features:

$$\tilde{\mathbf{z}} = \sum_b \alpha_b \mathbf{z}_b \quad (6)$$

This vector is then passed through a linear projection layer to produce the final output logit $\mathbf{s}_{fuse}$. Through this process, the

model adaptively assigns higher weights to modality pairs that are more informative for clickbait detection. The total loss is computed by combining branch-level logits and the fused logit using weighted binary cross-entropy (BCE) loss.

$$\mathcal{L} = \lambda_{br} \sum_b \text{BCE}(s_b, \hat{y}) + \mu \text{BCE}(s_{fuse}, \hat{y}) \quad (7)$$

Here, $\hat{y}$ denotes the target label with smoothing applied. During inference, the sigmoid function is applied to the fused logit, and $\sigma(s_{fuse})$ is used as the final prediction probability. The resulting branch attention weights α provide a quantitative measure of how much each modality pair (e.g., title–STT, thumbnail caption–keyframe captions) contributes to the clickbait prediction.

4. Experiment

4.1 Setup

Data

For training and evaluating the proposed model, we constructed a dataset consisting of 800 multimodal samples following the procedure described in Section 2. The dataset was balanced, comprising 400 clickbait and 400 non-clickbait samples. Among the clickbait samples, 200 were generated by swapping the titles and thumbnails between videos with similar topics, while the remaining 200 were generated using the AI-based method. The non-clickbait samples consisted of 200 original news videos and 200 augmented samples generated by paraphrasing the original titles while preserving their semantic meaning. The 800 samples were divided into training, validation, and test sets containing 560, 120, and 120 samples, respectively. Each sample consists of four modalities: title, thumbnail caption, STT, and keyframe captions, each of which was embedded using SBERT. In the proposed model, four semantic comparison pairs were constructed: title–STT (TS), title–keyframe (TK), thumbnail–STT (ThS), and thumbnail–keyframe (ThK). These pairs were used to model the semantic relationships between the information conveyed by the title and thumbnail and the actual video content. In addition to the constructed dataset, we employed *YTClickbait21K* [11], a publicly available dataset consisting of real YouTube videos collected across diverse content domains, to evaluate the generalization ability of the proposed model in real-world environments. Considering the focus of this study on short video length, we selected videos approximately five minutes or shorter. Among these videos, 150 clickbait samples exhibiting semantic inconsistencies between the title or

thumbnail and the actual video content were selected, along with 150 non-clickbait samples, resulting in a balanced real-world test set of 300 samples. The real-world dataset was divided into a validation set of 100 samples and a test set of 200 samples. The validation set was used only to determine the decision threshold, while the test set was used for the final performance evaluation. None of the real-world samples were used for model training. This setting was used to examine whether the model trained on the constructed dataset could generalize to semantic inconsistency-based clickbait occurring in real-world YouTube content across diverse content domains.

Implementation details

We trained the model using AdamW with a learning rate of 1×10^{-3} and a weight decay of 1×10^{-3} . A ReduceLROnPlateau scheduler (factor = 0.5, patience = 3) was applied based on the validation balanced accuracy. Training was conducted for up to 80 epochs with a batch size of 32, and early stopping with a patience of 12 was employed. Each branch MLP has a hidden dimension of 64 with a dropout rate of 0.4. The training objective was defined as a weighted combination of the binary cross-entropy loss for the final fused prediction and the binary cross-entropy loss for the branch-level predictions, with weights of $\mu = 1.5$ and $\lambda_{br} = 0.5$ respectively. The decision threshold for classification was tuned on the validation set and subsequently applied to the corresponding test set. The same procedure was applied to the real-world evaluation. All experiments were independently repeated 20 times with different random seeds. The final results are reported as the mean and standard deviation (mean $\pm$ SD) across the 20 runs. Model performance was evaluated using Area Under the ROC Curve (AUC), Accuracy (ACC), Precision, Recall, and F1-score (F1).

4.2 Main Results

To evaluate the performance of the proposed method, we constructed five baselines employing different multimodal fusion and semantic relationship modeling strategies. **Independent Pairwise Branch** constructs four pairwise branches—TS, TK, ThS, and ThK—and processes each branch independently using an MLP classifier. The final prediction is obtained by averaging the branch-level prediction scores without explicitly integrating information across branches. **Unimodal Score Averaging** [5] independently processes the title, thumbnail caption, STT, and keyframe captions using separate MLP classifiers and obtains the final prediction by averaging the four modality-level prediction scores. Unlike the pairwise approaches, this baseline does not explicitly model cross-modal semantic relationships. **Cosine Similarity** uses the four pairwise cosine similarity scores (TS,

TK, ThS, and ThK) as input features for clickbait classification, examining whether explicit similarity scores alone are sufficient to capture semantic relationships. **Cross-Attention** represents the title, thumbnail caption, STT, and keyframe captions as modality tokens and applies a Transformer-based attention encoder to model interactions among them. The resulting multimodal representation is then used for final clickbait classification. **Causal Representation Adapted** [12] decomposes the multimodal representation into invariant and variant components and performs prediction primarily using the invariant representation, providing a robustness-oriented representation learning baseline. These baselines enable comparison with unimodal prediction, explicit pairwise modeling, similarity-based modeling, attention-based multimodal interaction, and robustness-oriented representation learning. For a fair comparison, all baselines and the proposed method used the same four modalities: title, thumbnail, STT, and keyframe. The same data splits and evaluation settings were also applied to all methods, allowing performance differences to be attributed to differences in modality representation and fusion strategies rather than to differences in the input information.

4.2.1 Result on Synthetic Data

Table 1. Comparison of fusion strategies in Synthetic Data

Method	AUC	ACC	F1
Independent-Pairwise Branch	0.775 ± 0.006	0.717 ± 0.014	0.703 ± 0.026
Unimodal Score Averaging	0.745 ± 0.019	0.671 ± 0.056	0.706 ± 0.022
Cosine-Similarity	0.774 ± 0.016	0.696 ± 0.039	0.656 ± 0.086
Cross-Attention	0.748 ± 0.023	0.696 ± 0.033	0.685 ± 0.026
Causal Representation (adapted)	0.784 ± 0.009	0.723 ± 0.021	0.718 ± 0.027
Pairwise-Semantic Fusion (ours)	0.857 ± 0.018	0.735 ± 0.038	0.766 ± 0.024

Table 1 reports the Area Under the ROC Curve (AUC), Accuracy (ACC), and F1-score (F1) for each fusion strategy, with results presented as the mean $\pm$ standard deviation across 20 independent runs. We validate the effectiveness of the proposed pairwise semantic approach through comparison with the Independent-Modality Fusion method commonly used in prior studies. Pairwise-Semantic Fusion consistently outperforms the Independent-Modality Fusion baseline across all three metrics, achieving an AUC of 0.857, an accuracy of 0.735, and an F1-score of 0.766. These results indicate that learning relational

representations across modalities effectively captures semantic inconsistency, which is the core factor in clickbait detection.

4.2.2 Result on Real World Data

Table 2. Comparison of fusion strategies in Real-World Data

Method	AUC	ACC	F1
Independent-Pairwise Branch	0.713 ± 0.041	0.634 ± 0.039	0.592 ± 0.092
Unimodal Score Averaging	0.713 ± 0.038	<u>0.637 ± 0.035</u>	0.647 ± 0.103
Cosine-Similarity	0.824 ± 0.085	0.755 ± 0.079	<u>0.753 ± 0.078</u>
Cross-Attention	0.713 ± 0.054	0.640 ± 0.039	0.641 ± 0.057
Causal Representation (adapted)	0.672 ± 0.035	0.617 ± 0.032	0.602 ± 0.048
Pairwise-Semantic Fusion (ours)	<u>0.851 ± 0.037</u>	<u>0.757 ± 0.033</u>	<u>0.766 ± 0.048</u>

Table 2 presents the performance comparison of the different methods on real-world YouTube data. The proposed **Pairwise-Semantic Fusion** achieved the highest average performance across all evaluation metrics, with an AUC of 0.851, ACC of 0.757, and F1-score of 0.766. Among the baselines, **Cosine Similarity** achieved the best performance, with an AUC of 0.824, ACC of 0.755, and F1-score of 0.753. These results indicate that explicitly leveraging pairwise semantic relationships between the title and thumbnail and the actual video content is effective for real-world clickbait detection. Beyond using simple cosine similarity scores, the proposed method learns representations of individual modality-pair relationships and adaptively incorporates the contribution of each branch, resulting in further improvements in AUC and F1-score. In contrast, **Unimodal Score Averaging**, which processes each modality independently, and **Cross-Attention**, which models interactions among modalities through a general attention mechanism, both achieved an AUC of 0.713. Similarly, **Independent Pairwise Branch**, which constructs pairwise branches but simply averages their prediction scores, achieved an AUC of 0.713. These results suggest that, for real-world data, effectively representing and integrating the semantic relationships of individual modality pairs is important beyond unimodal prediction or general multimodal interaction modeling. Furthermore, the proposed model was trained solely on the constructed dataset, with no real-world samples used for model training. Nevertheless, its strong performance on real-world YouTube data suggests that the pairwise semantic representations learned from the Synthetic dataset can generalize to clickbait detection in real-world

environments.

4.3 Analysis

4.3.1 Branch-wise Attention Contribution

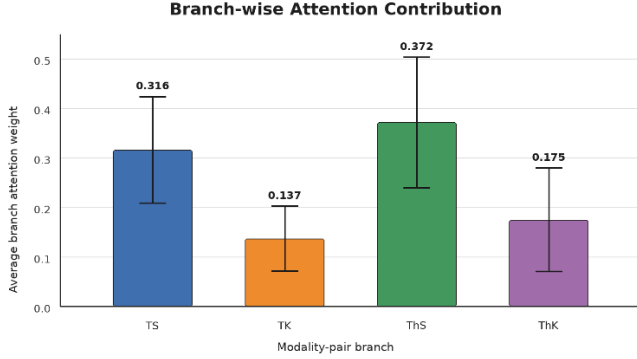


Figure 4. Average branch-level attention weights for each modality pair

We analyzed the branch-level attention weights of four modality-pair branches: TK (title–keyframe), TS (title–STT), ThK (thumbnail–keyframe), and ThS (thumbnail–STT). Figure 4 presents the mean attention weights across 20 independent runs, with error bars representing the standard deviation. The ThS branch exhibits the highest average attention weight (0.372), followed by TS (0.316), while TK (0.137) and ThK (0.175) show relatively lower contributions. These results indicate that modality pairs involving STT provide stronger signals for detecting semantic inconsistency, while keyframe-based information plays a complementary role.

4.3.2 Ablation Study

Table 3. Performance comparison of different pairwise branch combinations

Modality Pair	AUC	ACC	F1
TS	0.829 ± 0.011	0.750 ± 0.024	0.758 ± 0.014
ThS	0.843 ± 0.028	0.737 ± 0.041	0.764 ± 0.021
TK	0.785 ± 0.015	0.647 ± 0.026	0.720 ± 0.015
ThK	0.766 ± 0.028	0.647 ± 0.027	0.704 ± 0.018
TS+ThS	0.864 ± 0.010	0.743 ± 0.056	<u>0.771 ± 0.024</u>
TS+TK	0.829 ± 0.010	0.739 ± 0.026	0.751 ± 0.016
TK+ThK	0.793 ± 0.019	0.665 ± 0.021	0.725 ± 0.012
ThS+ThK	0.847 ± 0.008	<u>0.754 ± 0.019</u>	0.770 ± 0.012
ThS+TK+ThK	0.845 ± 0.013	0.755 ± 0.030	0.773 ± 0.019
TS+ThS+ThK	<u>0.858 ± 0.014</u>	0.732 ± 0.042	0.765 ± 0.017
TS+TK+ThK	0.829 ± 0.016	0.716 ± 0.045	0.747 ± 0.017
TS+ThS+TK	0.856 ± 0.021	0.725 ± 0.048	0.766 ± 0.028
All Branches	0.857 ± 0.018	0.735 ± 0.038	0.766 ± 0.024

To investigate the impact of different modality-pair combinations on clickbait detection performance, we conducted an ablation study by varying the branch configurations, as summarized in Table 3. Among the single-branch configurations, TS and ThS generally outperform TK and ThK. In particular, ThS achieves the highest AUC (0.843) and F1-score (0.764) among the single branches, while TS achieves the highest accuracy (0.750). This trend suggests that branches involving STT provide relatively informative cues for modeling semantic relationships between the title or thumbnail and the actual video content.

Combining multiple branches can further improve performance, although the improvement varies depending on the branch configuration. Among the two-branch configurations, TS+ThS achieves the highest AUC of 0.864, while ThS+ThK achieves an accuracy of 0.754 and an F1-score of 0.770. These results indicate that combining complementary semantic relationships can improve performance compared with relying on individual branches. However, adding more branches does not consistently lead to further improvements. For example, the three-branch configuration ThS+TK+ThK achieves the highest accuracy (0.755) and F1-score (0.773), whereas other three-branch configurations show lower performance on some metrics.

The All Branches configuration achieves an AUC of 0.857, an accuracy of 0.735, and an F1-score of 0.766. Although it does not achieve the highest performance on every metric, it maintains competitive performance by jointly incorporating all four semantic relationships. Overall, these results indicate that performance depends not only on the number of branches but also on **which modality-pair relationships are combined**, highlighting the importance of effectively integrating complementary semantic information.

4.3.3 Comparison of Pooling Methods

Table 4. Performance comparison of different pooling strategies

Pooling	AUC	ACC	F1
Mean	0.824 ± 0.026	0.744 ± 0.049	0.755 ± 0.026
Top-k	0.819 ± 0.023	0.704 ± 0.055	0.741 ± 0.024
Top-k(sim)	<u>0.848 ± 0.017</u>	<u>0.736 ± 0.041</u>	0.770 ± 0.022
Attention	0.857 ± 0.018	0.735 ± 0.038	<u>0.766 ± 0.024</u>

This study compares four pooling strategies to investigate how different sequence summarization methods affect model performance: mean pooling (element-wise averaging) [10], Top-k pooling (magnitude-based selection) [13], Top-k(sim) pooling (similarity-based selection) [14], and attention pooling (query-aware weighting) [9]. Table 4 presents the performance comparison across these methods. Among the four strategies, **attention pooling achieved the highest AUC of 0.857 ± 0.018** ,

indicating the strongest overall discriminative ability. **Top-k(sim)** also showed competitive performance, achieving an AUC of 0.848 ± 0.017 and the highest F1-score of 0.770 ± 0.022 . In contrast, **mean pooling achieved the highest ACC of 0.744 $\pm$ 0.049**, although its AUC was lower at 0.824 ± 0.026 . Top-k pooling showed relatively lower performance, with an AUC of 0.819 ± 0.023 and an F1-score of 0.741 ± 0.024 . The relatively strong performance of attention pooling and Top-k(sim) suggests that incorporating query-dependent semantic relevance can be beneficial when aggregating variable-length STT and keyframe representations. Unlike Top-k(sim), which retains only a subset of elements based on their similarity to the query, attention pooling assigns weights across the entire sequence, allowing information from all elements to be preserved while emphasizing those that are more relevant to the query. Overall, these results indicate that **query-aware pooling provides an effective strategy for modeling semantic relationships between the title or thumbnail and the corresponding video content**, although the relative advantage varies depending on the evaluation metric.

5. Conclusion

Conflict of Interest

No potential conflict of interest relevant to this article was reported.

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